Diagrammatical explanation of marginal cost [*MC*]

Marginal cost is the change in aggregate cost when the volume of production is increased or decreased by one unit.

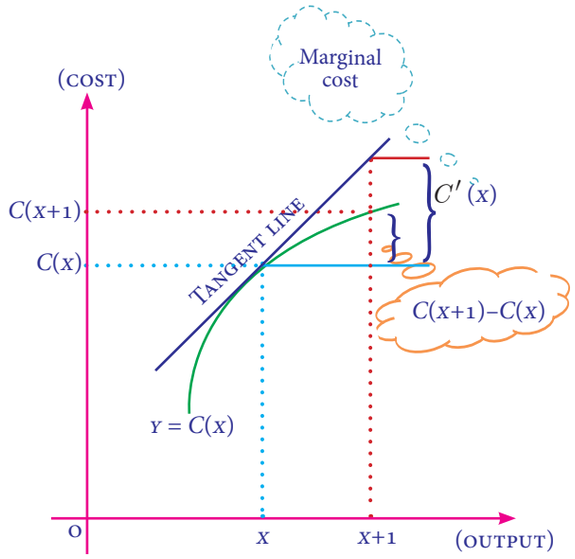


Fig: 6.5

Revenue function

Revenue is the amount realised on a commodity when it is produced and sold. If x is the number of units produced and sold and p is its unit price, then the total revenue function $R(x)$ is defined as $R(x) = px$, where x and p are positive.



The revenue increases when the Producer, supplies more quantity at a higher price.

Some standard results

If $R(x) = px$ be the revenue function, then

(i) Average revenue:

$$AR = \frac{Total\ Revenue}{Output} = \frac{R(x)}{x} = p$$

(ii) Marginal revenue:

$$MR = \frac{dR}{dx} = \frac{d}{dx}(R(x)) = R'(x)$$

(iii) Marginal average revenue:

$$MAR = \frac{d}{dx}(AR) = AR'(x)$$

Remarks:

(i) Average revenue [AR] and price [p] are the same. [i.e. $AR=p$]

(ii) The marginal revenue [MR] is approximately equal to the additional revenue made on selling of $(x+1)^{th}$ unit, when the sales level is x units.

Diagrammatical explanation of Marginal Revenue [*MR*]

Marginal revenue is the change in aggregate revenue when the volume of selling unit is increased by one unit.

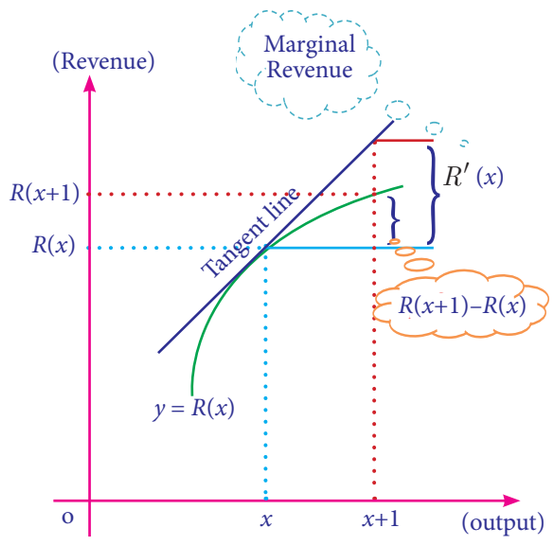


Fig : 6.6

Profit function

The excess of total revenue over the total cost of production is called the profit. If $R(x)$ is the total revenue and $C(x)$ is the total cost, then profit function P(x) is defined as P(x) = $R(x)$ – $C(x)$

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